Multi-Object Detection and Persistent ID Tracking

1. Introduction

This project focuses on building a computer vision pipeline for detecting and tracking multiple objects in public sports/event video footage. The goal is to assign a unique and consistent ID to each detected subject and maintain that identity across the entire video, even under challenging real-world conditions.

2. Approach Overview

The system follows a sequential pipeline:

- Input video is processed frame by frame
- Object detection is performed on each frame
- Detected objects are passed to a tracking algorithm
- Unique IDs are assigned and maintained across frames
- The final output video is annotated with bounding boxes and IDs

3. Object Detection

For object detection, YOLOv8 (Ultralytics) was used.

Why YOLOv8?

- High speed suitable for real-time applications
- Good detection accuracy
- Easy integration with Python and OpenCV

4. Multi-Object Tracking

For tracking, DeepSORT was used.

Why DeepSORT?

- Combines motion and appearance features
- Maintains identity even during partial occlusion
- Works well with real-time detection pipelines

DeepSORT takes bounding boxes from YOLO and assigns a unique ID to each detected subject. It uses:

- Kalman Filter for motion prediction
- Appearance embeddings for identity matching

5. ID Consistency Strategy

To maintain consistent IDs across frames:

- Tracker parameters such as max_age and cosine distance were tuned
- Low-confidence detections were filtered out
- Frame skipping was applied to reduce noise and improve stability

These improvements helped reduce ID switching and improved tracking performance.

6. Additional Features Implemented

To enhance the system, the following features were added:

- Trajectory visualization: Displays movement path of each tracked subject
- Player count: Shows number of active tracked subjects
- FPS monitoring: Displays real-time performance of the system
- Using advanced re-identification models for better ID stability
- Deploying as a web application (e.g., Streamlit)

10. Conclusion

This project successfully demonstrates a complete pipeline for multi-object detection and persistent ID tracking in sports/event footage. It handles real-world challenges reasonably well and provides a strong foundation for further enhancements in sports analytics and surveillance systems.